6C STANDARD ENTHALPY CHANGE OF COMBUSTION

SPECIFICATION REFERENCE: 6.4 PART, 6.5, 6.8 PART

LEARNING OBJECTIVES

- Be able to define standard enthalpy change of combustion.
- Understand simple experiments to measure enthalpy changes in terms of evaluating sources of error and assumptions made in the experiments.
- Be able to process results from experiments to calculate a value for the enthalpy change of combustion of a substance.

WHAT IS MEANT BY STANDARD ENTHALPY CHANGE OF COMBUSTION

The **standard enthalpy change of combustion** ($\Delta_c H^\ominus$) is the enthalpy change measured at 100 kPa and a specified temperature, usually 298 K, when one mole of a substance is completely burned in oxygen.

When you are writing an equation to represent the standard enthalpy change of combustion, it is important that you specify it is one mole of the substance that is being burned. Two common ways of writing the combustion of hydrogen are:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \text{ and}$$

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

The first equation, in which one mole of hydrogen undergoes combustion, represents $\Delta_c H^\ominus$. The enthalpy change for the second equation is $2 \times \Delta_c H^\ominus$.

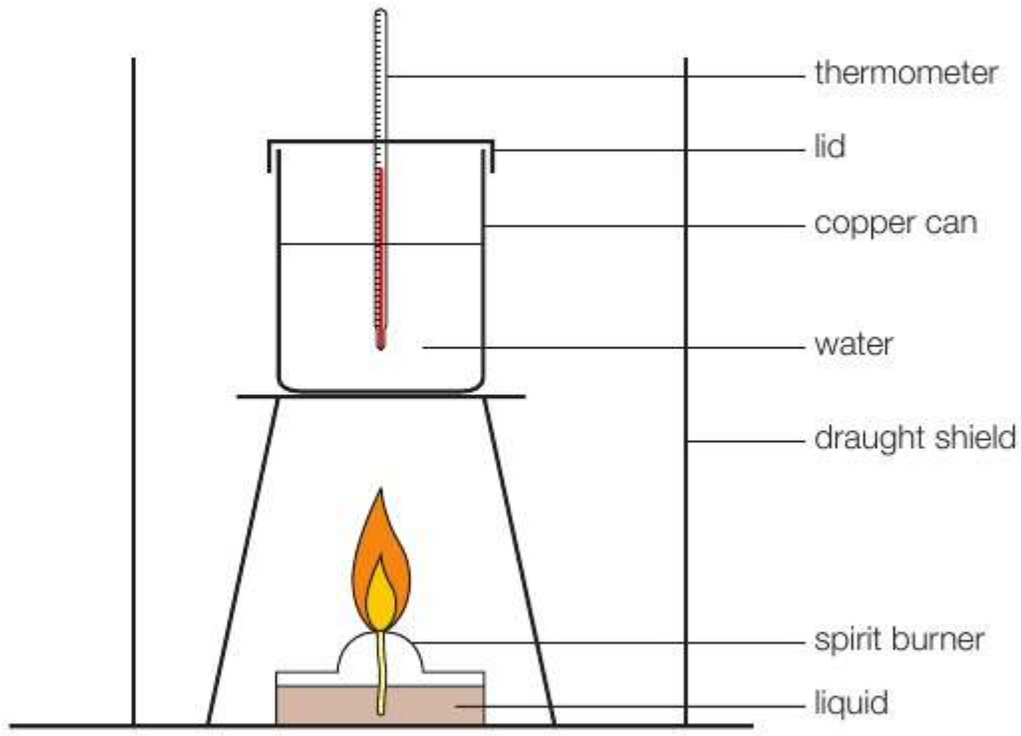
EXPERIMENTAL DETERMINATION OF ENTHALPY CHANGE OF COMBUSTION OF A LIQUID

To find the enthalpy change of combustion of a liquid, a known mass of the liquid is burned and the heat energy released is used to heat a known volume of water.

The following procedure is used.

- A spirit burner containing the liquid being tested is weighed.
- A known volume of water is added to a copper can.
- The temperature of the water is measured.
- The burner is lit.
- The mixture is constantly stirred with the thermometer.
- When the temperature of the water has reached approximately 20 °C above its initial temperature, the flame is extinguished and the burner is immediately reweighed.
- The final temperature is measured.

The appropriate laboratory apparatus is shown in **fig A**.



▲ **fig A** Laboratory apparatus to find the enthalpy change of combustion of a liquid.

A typical set of results for ethanol (C_2H_5OH, molar mass 46.0 g mol^{-1}) are shown in **table A**.

volume of water heated	100.0 cm^3
mass of ethanol burned	0.420 g
temperature change, ΔT	+24.5 °C

table A

CALCULATING ENTHALPY CHANGE OF COMBUSTION

The enthalpy change of combustion is now calculated in three stages.

Stage 1: Calculate the heat energy, Q, transferred to the water using the equation

$Q = mc\Delta T$, where m is the mass of water and c is the specific heat capacity of water.

If m is in grams, then c is quoted as 4.18 J g^{-1} K^{-1}.

Assuming the density of water is 1.00 g cm^{-3}, then m = 100.0 g.

The temperature *change*, ΔT, has the same value in K as it does in °C.

So, $Q = 100.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times +24.5\,K$

$= +10\,241\,J = +10.24\,kJ$

Stage 2: Calculate the amount, n, of ethanol burned.

$$n(C_2H_5OH) = \frac{0.420\,g}{46.0\,g\,mol^{-1}} = 9.13 \times 10^{-3}\,mol$$

Stage 3: Calculate $\Delta_c H^\ominus$, using the equation

$$\Delta H = -\frac{Q}{n}$$

$$\Delta_c H^\ominus = -\frac{+10.24\,kJ}{9.13 \times 10^{-3}\,mol} = -1120\,kJ\,mol^{-1}$$

(to 3 significant figures)

It is very important that you include a sign with any value of ΔH that you quote.

LEARNING TIP

When the two equations from stages 1 and 3 are used together, they result in the correct sign for ΔH. In this example, ΔH is negative, which is consistent with the exothermic reaction that is taking place.

If the temperature had decreased during the reaction, then ΔT would be negative, which in turn would make Q negative. This would lead to a positive value for ΔH, consistent with an endothermic reaction.

If in doubt, always use your common sense. If the temperature increases, the reaction must be exothermic, and vice versa.

EVALUATING SOURCES OF ERROR AND ASSUMPTIONS MADE IN THE EXPERIMENTS

The value obtained from the above experiment is in reasonable agreement with the standard enthalpy change of combustion of ethanol, as obtained from a data book, of $-1367\,kJ\,mol^{-1}$. This means that the errors in procedure were minimal.

Here are some possible sources of error.

- Some of the heat energy produced in burning is transferred to the air and not the water.
- Some of the ethanol may not burn completely to form carbon dioxide and water. (Incomplete combustion would produce less heat energy and also cause soot to form on the bottom of the copper can.)
- Some of the heat energy produced in burning is transferred to the copper can and not to the water.
- The conditions are not standard. For example, water vapour, not liquid water, is produced.
- The experiment takes a long time. This means that not all of the heat energy transferred from the water to the surroundings is compensated for.

CHECKPOINT

1. Ethanol and methoxymethane have the same molecular formula, C_2H_6O.

The standard enthalpy change of combustion at 298 K of ethanol gas and methoxymethane (CH_3OCH_3) gas are $-1367\,kJ\,mol^{-1}$ and $-1460\,kJ\,mol^{-1}$ respectively.

(a) Write an equation to represent the standard enthalpy change of combustion of: (i) ethanol, and (ii) methoxymethane.

(b) Suggest why the two compounds have different standard enthalpy changes of combustion despite having the same molecular formula.

2. The table shows the results of separately combusting 1.00 g of each of four alcohols and determining the amount of energy required to produce the same temperature rise in each reaction.

SKILLS ADAPTIVE LEARNING

ALCOHOL	MOLAR MASS / $g\,mol^{-1}$	ENERGY REQUIRED / $kJ\,g^{-1}$	ENERGY REQUIRED / $kJ\,mol^{-1}$
methanol	32.0	22.34	
ethanol	46.0	29.80	
propan-1-ol	60.0	33.50	
butan-1-ol	74.0	36.12	

(a) Complete the table by calculating the energy required per mole for each of the four alcohols. Quote the values for $\Delta_c H$ for each of them.

(b) Draw a graph of $\Delta_c H$ (vertical axis) against the number of carbon atoms in one molecule and use it to estimate a value for the enthalpy change of combustion of pentan-1-ol under the same conditions.

(c) Extrapolate the graph line to 0 on the x-axis and comment on the value of $\Delta_c H$ you read off from the graph.

EXAM HINT

To *extrapolate* means to extend the graph on the basis that the trend shown in the points recorded continues. So, if the points recorded suggest a curve then you should attempt to continue the curve.

SUBJECT VOCABULARY

standard enthalpy change of combustion ($\Delta_c H^{\ominus}$) the enthalpy change measured at 100 kPa and a stated temperature, usually 298 K, when one mole of a substance is completely burned in oxygen